Quiz — Assignment 3

ECON 5783 — University of Arkansas

Name: _____

1. Suppose that, conditional on age, smoking during pregnancy is randomly assigned:

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp \text{Mother Smokes}_i \mid \text{Age}_i.$$

   (a) Write the conditional average treatment effect for 30-year-old mothers using observable conditional means. Then explain how you would combine age-specific treatment-effect estimates to estimate the overall average treatment effect in the sample.

   (b) Give a concrete example of why you think the conditional independence assumption might fail in this setup.